

姓名：廖汶鋒

(號)

成績

年級：微 91

科目：Probability and Stochastic Processes (I)

1. (a) Denote the events

 $A = \{\text{The source transmits a 0}\}$ $B = \{\text{The source transmits a 1}\}$ $C = \{A \text{ 0 is received}\}$ $D = \{A \text{ 1 is received}\}$ As we know, $A \cap B = \emptyset$, $A \cup B = \Omega$, $P(A) = p$, $P(B) = 1-p$ $P(C|A) = 1 - \epsilon_0$, $P(D|A) = \epsilon_0$ $P(C|B) = \epsilon_1$, $P(D|B) = 1 - \epsilon_1$

$$\begin{aligned}
 P(\text{Kth symbol is received correctly}) &= P(A \cap C) + P(B \cap D) \\
 &= P(C|A)P(A) + P(D|B)P(B) \\
 &= (1 - \epsilon_0)p + (1 - \epsilon_1)(1 - p)
 \end{aligned}$$

$$\begin{aligned}
 (b) P(1011 \text{ is received correctly}) &= P(1011 \text{ is transmitted} \mid 1011 \text{ is received}) \\
 &= \frac{P(BAD)^3 P(AAC)}{P(BAD)^3 P(AAC)} = P(D|B)^3 P(C|A) \\
 &= \frac{(1 - \epsilon_1)^3 (1 - p)^3 (1 - \epsilon_0)p}{(1 - \epsilon_1)^3 (1 - p)^3 (1 - \epsilon_0)p} = (1 - \epsilon_1)^3 (1 - \epsilon_0)
 \end{aligned}$$

$$\begin{aligned}
 (c) P(A \text{ 0 is correctly decoded}) &= P(\text{Two 0s are received} \mid \text{"000" is transmitted}) \\
 &\quad + P(\text{Three 0s are received} \mid \text{"000" is transmitted}) \\
 &= \binom{3}{1} (1 - \epsilon_0)^2 \epsilon_0 + (1 - \epsilon_0)^3 = (1 - \epsilon_0)^2 (2\epsilon_0 + 1)
 \end{aligned}$$

$$\frac{(1 - \epsilon_0)^2 (2\epsilon_0 + 1)}{(1 - \epsilon_0)^2 (2\epsilon_0 + 1)} = \frac{[(1 - \epsilon_0) + (1 - \epsilon_0) + (2\epsilon_0 + 1)]^3}{3^3} = 1 \quad (\text{if and only if } 1 - \epsilon_0 = 2\epsilon_0 + 1, \text{ i.e. } \epsilon_0 = 0)$$

~~100~~

$$(d) P(a \text{ 0 is correctly decoded} \mid \text{"000" is transmitted}) \geq P(a \text{ 0 is correctly decoded} \mid a \text{ 0 is transmitted})$$

$$\Leftrightarrow (1 - \epsilon_0)^2 (2\epsilon_0 + 1) \geq 1 - \epsilon_0$$

$$\Leftrightarrow (1 - \epsilon_0)(-2\epsilon_0^2 + \epsilon_0) \geq 0 \Leftrightarrow (1 - \epsilon_0)\epsilon_0(1 - 2\epsilon_0) \geq 0$$

$$\Leftrightarrow 0 \leq \epsilon_0 \leq \frac{1}{2} \text{ or } \epsilon_0 = 1$$

$$\text{Also, } \epsilon_0(1 - \epsilon_0) \neq 0, \text{ therefore } 0 < \epsilon_0 \leq \frac{1}{2}$$

(e) $P(\text{a } \cancel{0} \text{ is transmitted} \mid \text{"000" is transmitted} \mid \text{"101" is } \overset{\text{received}}{\cancel{\text{transmitted}}})$

$$= P(A|D)^2 P(A|C)$$

$$= \left[\frac{P(D|A)P(A)}{P(D)} \right]^2 \left(\frac{P(C|A)P(A)}{P(C)} \right) = \frac{\epsilon^2 p}{\epsilon^2 p + (1-\epsilon)(1-p)}$$

$$= \left[\frac{P(D|A)P(A)}{P(D|A)P(A) + P(D|B)P(B)} \right]^2 \left[\frac{P(C|A)P(A)}{P(C|A)P(A) + P(C|B)P(B)} \right]$$

$$= \left[\frac{\epsilon p}{\epsilon p + (1-\epsilon)(1-p)} \right]^2 \left(\frac{(1-\epsilon)p}{(1-\epsilon)p + \epsilon(1-p)} \right)$$

2. $P_T = P(\dots)$

Denote the events

$R = \{\text{the score 4 to 6 in favor of Wendy}\}$

$A = \{\text{Telis wins on a hole}\}, P(A) = p$

$B = \{\text{Wendy wins on a hole}\}, P(B) = (1-p)$

$$P_T = P(\text{Telis wins not less than 6 times from 11th hole to 18th hole} \mid R)$$

$$= \binom{8}{6} p^6 (1-p)^2 + \binom{8}{7} p^7 (1-p) + \binom{8}{8} p^8$$

$$= p^6 [28(1-p)^2 + 8p(1-p) + p^2] = p^6 (21p^2 - 48p + 28)$$

$$P_W = P(\text{Wendy wins not less than 4 times from 11th hole to 18th hole} \mid R)$$

$$= \binom{8}{4} p^4 (1-p)^4 + \binom{8}{5} p^5 (1-p)^3 + \binom{8}{6} p^6 (1-p)^2 + \binom{8}{7} p^7 (1-p) + \binom{8}{8} (1-p)^8$$

$$= 1 - P_T - \binom{8}{3} p^5 (1-p)^3$$

$$\Leftrightarrow P_W + P_T = 1 - 56p^5(1-p)^3$$

Therefore, Telis should get $\frac{10p^6(21p^2 - 48p + 28)}{1 - 56p^5(1-p)^3}$

姓名：

(號)

成績

年級：

科目：

~~proof: $P(A \cap B | C) = P(A | C) P(B | C)$~~

proof: (1) First, we proof " $P(A \cap B | C) = P(A | C) P(B | C) \nRightarrow P(A \cap B) = P(A) P(B)$ "

~~Denote~~ $A = \{$

Consider rolling a fair 4-sided die, twice

Denote the event

$$A = \{\text{1st result in 1}\}$$

$$B = \{\text{the sum of the results is 4}\}$$

$$C = A \cap B$$

Meanwhile, $P(A \cap B | C) = P(A | C) = P(B | C) = 1$

But $P(A \cap B) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

$$P(A) = \frac{1}{4}$$

$$\Rightarrow P(A \cap B) \neq P(A) P(B)$$

$$P(B) = \frac{1}{4} \times \frac{1}{4} \times 3 = \frac{3}{16}$$

(2) Second, we proof " $P(A \cap B) = P(A) P(B) \nRightarrow P(A \cap B | C) = P(A | C) P(B | C)$ "

We roll a fair 4-sided die twice again

Denote the event

$$A = \{\text{1st result in 1}\}$$

$$B = \{\text{2nd result in 2}\}$$

$$C = \{\text{the sum of result is 4}\}$$

Obviously, we have $P(A \cap B) = P(A) P(B)$

But $P(A \cap B | C) = 0$

$$P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{1}{4} \times \frac{1}{4}}{\frac{3}{16}} = \frac{1}{3}$$

$$P(B | C) = \frac{P(B \cap C)}{P(C)} = \frac{\frac{1}{16}}{\frac{3}{16}} = \frac{1}{3}$$

$$\Rightarrow P(A \cap B | C) \neq P(A | C) P(B | C)$$

姓名：

(號)

成績

年級：

科目：

(3) Now, we find an event C which satisfies

$$\forall A, B \in \Omega, P(A \cap B) = P(A)P(B) \Leftrightarrow P(A \cap B | C) = P(A|C)P(B|C)$$

$$\text{Let } A = C, B = C^c$$

$$P(A \cap B | C) = P(C \cap C^c | C) = P(\emptyset | C) = 0$$

$$P(A|C) = 1, P(B|C) = P(C^c | C) = 0.$$

$$\Rightarrow P(A \cap B | C) = P(A|C) \times P(B|C)$$

$$\text{Therefore, } C \text{ satisfies, } P(C \cap C^c) = P(C)P(C^c) = P(C)[1 - P(C)]$$

$$\Leftrightarrow P(C)[1 - P(C)] = 0$$

$$\Leftrightarrow P(C) = 0 \text{ or } P(C) = 1$$

$$\Leftrightarrow C = \emptyset \text{ or } C = \Omega$$

As C cannot be the empty set, C can only be the whole sample space Ω .

$$4. (a) P(A) = 2 \times \left(\frac{1}{2}\right)^3 = \frac{1}{4}, P(B) = \binom{3}{1} \times \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 = \frac{1}{2}, P(C) = 1 - P(A^c) = 1 - P(A) = \frac{3}{4}$$

$$P(A \cap B) = P(\text{all children are girls}) = \left(\frac{1}{2}\right)^3$$

$$P(B \cap C) = P(1 \text{ boy and } 2 \text{ girls}) = \binom{3}{1} \times \left(\frac{1}{2}\right)^3 = \frac{3}{8}$$

$$\text{Now, we have } P(A \cap B) = P(A)P(B) \text{ and } P(B \cap C) = P(B)P(C)$$

$\Leftrightarrow A$ and B are independent, B and C are independent.

$$(b) P(A \cap C) = P(\emptyset) = 0 \neq P(A)P(C) \Rightarrow A \text{ and } C \text{ aren't independent}$$

(c) Assume the ~~total~~ probability that a child is a boy is $p_b = p$
the probability that a child is a girl is $p_g = 1 - p_b = 1 - p$

$$P(A) = p^3 + (1-p)^3, P(B) = \binom{3}{1} p(1-p)^2 + (1-p)^3, P(C) = P(A^c) = 1 - p^3 - (1-p)^3$$

$$P(A \cap B) = (1-p)^3, P(B \cap C) = \binom{3}{1} p(1-p)^3 = 3p(1-p)^3, P(A \cap C) = 0$$

$$P(A)P(B) = (1-3p+3p^2)(1-p)^2(2p+1) \neq P(A \cap B)$$

$$P(B)P(C) = (1-p)^2(2p+1)3p(1-p) = 3p(1-p)^3(2p+1) \neq P(B \cap C)$$

$$P(A)P(C) = (1-3p+3p^2) \cdot 3p(1-p) \neq P(A \cap C) \text{ [if } p=0 \text{ or } p=1]$$

$\Rightarrow$ The results above may not hold.

$$(d) P(A) = 2 \times \left(\frac{1}{2}\right)^4 = \frac{1}{8}, \quad P(B) = \binom{4}{1} \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 = \frac{5}{16}, \quad P(C) = 1 - P(A) = \frac{7}{8}$$

$$P(A \cap B) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}, \quad P(B \cap C) = P(1 \text{ boy and } 3 \text{ girls}) = \binom{4}{1} \left(\frac{1}{2}\right)^4 = \frac{1}{4}$$

$$P(C \cap A) = 0$$

$$\Rightarrow P(A \cap B) \neq P(A)P(B), \quad P(B \cap C) \neq P(B)P(C), \quad P(C \cap A) \neq P(C)P(A)$$

$\Leftrightarrow A, B, C$ are independent pairwise.